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Axially slanted laser illumination scheme for direct and accurate Raman spectroscopic determination of gemcitabine concentration in freeze-dried gemcitabine injection powder housed in a glass container

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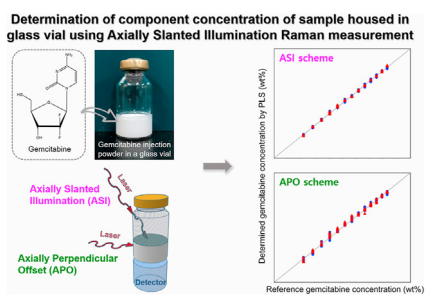
HIGHLIGHTS

- Axially slanted illumination (ASI) scheme was proposed for sampling-free Raman analysis.
- ASI scheme slantingly irradiated laser on headspace side of a sample-housed glass container.
- Using ASI scheme, gemcitabine injection powder in a glass container was directly measured.
- Glass peak was reduced by the longer distance between laser illumination spot and detector.
- Accuracy of compositional analysis improved due to the minimized influence of glass peak.

GRAPHICAL ABSTRACT



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ABSTRACT

When Raman spectroscopy is employed for a direct *in situ* determination of ingredient concentration for a product stored in a glass container, minimization of the interfering glass background in the collected spectrum is demanding to secure a more accurate analysis. To meet this request, an axially slanted illumination (ASI) scheme slantingly irradiating laser on the headspace side of a glass container and positioning a detector beneath the container was demonstrated in this study. This ASI scheme was basically designed to increase the distance between the laser illumination spot and detector location to minimize the number of glass photons reaching the detector. The analytical utility of the scheme was evaluated for the determination of gemcitabine concentration (42.9–58.2 wt%) in the gemcitabine injection powder housed in a glass container. Using the ASI scheme, the spectral features of the gemcitabine powder became distinct with only a weak underlying glass background signal. For comparative purpose, when an axially perpendicular offset (APO) scheme perpendicularly irradiating laser on the side wall where the sample was filled was used, the magnitude of glass background was higher, and the most intense gemcitabine peak was largely buried in the glass peak. The accuracy for determination of gemcitabine concentration using the ASI scheme was superior with an error of 0.20 wt%, while 0.33 wt%

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with employing the APO scheme. Overall, this study demonstrates that the ASI scheme is a potentially versatile Raman spectroscopic tool for fast non-sampling analysis of other products stored in a glass container.

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1. Introduction

A fast, sampling-free direct spectroscopic analysis of finished products, stored in various containers such as glass and plastic bottles, for identification and quality control has become demanding, especially in the pharmaceutical, food, and security sectors [1–10]. Since Raman spectroscopy is superior to meet the demand, diverse studies on the through-glass container Raman measurements have been reported, for example, the detection of illicit drugs (heroin, cocaine, and amphetamine) [6] and counterfeit medical products [7], authentication of Scotch whisky [8], and identification of counterfeit alcohol [9] and fake spirits [10]. Meanwhile, the through-container measurement results in the co-appearance of sample and container peaks together in a collected spectrum; therefore, minimization or elimination of the container peak which could hamper observation of the analyte peak(s) with simultaneous maximization of the analyte peak intensities are key elements to secure the accuracy of the target analysis [4,5]. In this context, our research group demonstrated an axially perpendicular offset (APO) scheme [11–13], where the laser illumination and photon detection are oriented in an axially perpendicular geometrical arrangement, for the Raman spectroscopic analysis of an ethanol solution housed in different glass bottles. The APO scheme was effective in reducing the co-presenting broad glass peak, and the reduction was realized by avoiding the detection of the generated glass photons by positioning the detector farther from the laser illumination spot at an effective offset distance [12].

Meanwhile, if a sample housed in a glass container is a solid of highly scattering medium rather than a neat liquid like ethanol, the utility of the APO scheme for glass background reduction must be readdressed. When glass photons generated at a glass wall travel into a housed solid medium, they start to isotropically diffuse due to the scattering and the distribution of glass photons becomes predominantly broader compared to that in a neat solution [14–16]. Therefore, the APO scheme presents difficulty for avoidance of these glass photons in detection and complete suppression of the glass background is difficult to achieve. Therefore, an alternative spectral acquisition scheme able to minimize the presence of the glass background in a sample spectrum without loss of analyte information is further beneficial.

To address this issue in this study, direct determination of gemcitabine concentration in freeze-dried gemcitabine injection powder housed in a glass vial (refer to the picture in Fig. 1 (a)) has been attempted using Raman spectroscopy. Gemcitabine is a chemotherapy medication (cytotoxic agent) used to treat a number of different types of cancers [17–19]. Initially, the APO scheme (refer to Fig. 1 (b)) was employed to directly collect Raman spectra of the 51.8% gemcitabine powder in a glass vial and the magnitude of the glass background relative to gemcitabine peaks in the acquired spectra was investigated. Meanwhile, since the sample only partially fills the glass vial, an empty space is available inside (Fig. 1 (a)), an offset distance between locations of the laser irradiation and photon detection can be increased by setting the laser irradiation with a slant at the upper side of vial wall in the unoccupied headspace (refer to the axially slanted illumination (ASI) scheme, Fig. 1 (c)). Therefore, the ASI scheme would help to reduce the glass background while maintaining substantial intensity of the analyte peaks. The same sample was measured again using the ASI scheme and the subsequent spectral features were compared with those obtained using the APO scheme. Finally, Raman spectra of five batches of gemcitabine samples in glass vials were acquired using both the APO and ASI schemes, and the resulting accuracies for determination gemcitabine concentration using partial least squares (PLS) were compared.

2. Experimental section

2.1. Preparation of gemcitabine injection powder samples

Freeze-dried gemcitabine injection powder samples were prepared at Chong Kun Dang Pharmaceutical Co., Cheonan, Korea. Initially, the mixtures of gemcitabine·HCl (active pharmaceutical ingredient (API)), D-mannitol, and sodium acetate·$3H_2O$ were prepared. The concentrations of gemcitabine·HCl in the mixture samples varied from 42.9 wt% to 58.2 wt%. Each mixture sample was dissolved in water at a target pH and the resulting solution was filtered (polyethersulfone (PES) filter, pore size: 0.2 μm). Each filtered sample solution was transferred into a sterilized glass vial (USP Type I neutral borosilicate glass vial with a high hydrolytic resistance [20]) and lyophilized under the specified conditions. Then, the vial containing the lyophilized sample was capped. Thirteen samples with varying gemcitabine·HCl concentrations in the 42.9–58.2 wt% range were prepared and this procedure was separately repeated five times, yielding a total of 65 samples from five batches (13 samples per batch).

2.2. Acquisition of Raman spectra of the sample housed in a glass vial

In both APO and ASI measurements, a separated laser (Invictus, Kaiser, wavelength: 785 nm, 400 mW) was illuminated on the wall of the sample-housed glass vial to acquire the corresponding spectra using a Raman spectrometer (RamanRxn1 unit, Kaiser, Ann Arbor, MI). The diameter of the laser irradiation was approximately 1.0 mm. The detection probe (PhAT probe, Kaiser) was positioned beneath the vial bottom to collect the generated Raman photons. The detection diameter of the probe was 25.4 mm, nearly similar to that of the glass vial (25 mm). The pictures showing the real arrangement of laser and detector with respective to the sample-housed glass vial in the APO and ASI schemes are presented in the electronic supplementary material. Also, the schematic illustrations showing the passage of laser, glass, and sample photons in the APO and ASI schemes are included in the electronic supplementary material. Each spectrum was collected by accumulating 10 scans with a laser exposure time of 10 s at each scan (total spectral acquisition time: 100 s). Dark subtraction was performed before every spectral acquisition. For each sample, triplicate spectra were acquired by randomly illuminating the laser at different spots on the wall of the sample-containing vial. Raman spectra of the samples in each batch were acquired on different periods. Baseline correction, normalization of spectra, and PLS were conducted using MATLAB version 2017a (The Math Works Inc., Natick, MA, USA).

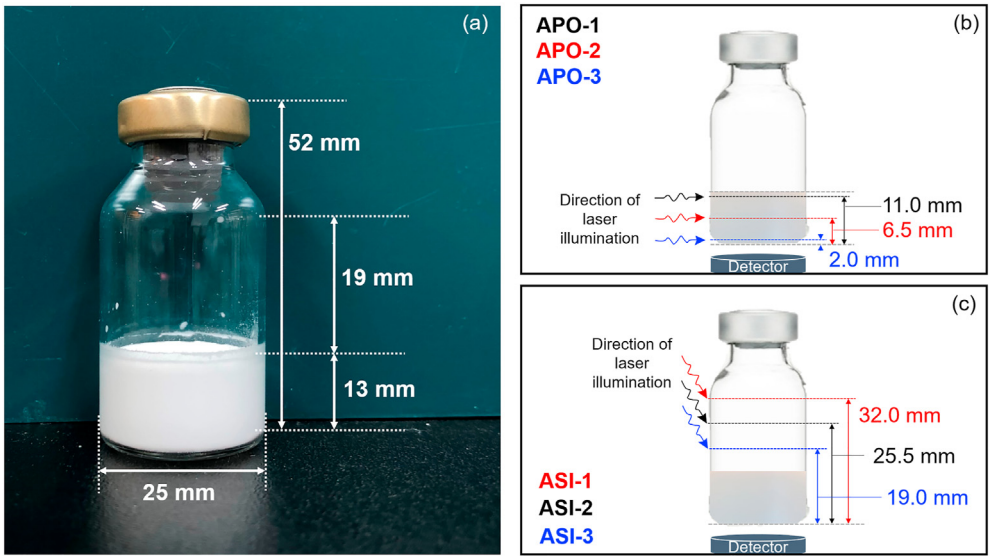

Fig. 1. Photograph of freeze-dried gemcitabine injection powder housed in a glass vial (a), and graphical descriptions of three different APO schemes (b) and ASI schemes (c) used to collect Raman spectra of the housed gemcitabine powder.

For PLS modeling, among the total five sample batches, three sample batches and the remaining two sample batches were assigned as the calibration (39 samples) and test (26 samples) sets, respectively. The ranges of gemcitabine concentration variation among the batches were same. The division of whole sample set by the batch was to examine potential batch-to-batch difference. The gemcitabine concentrations of all the samples employed in this study are designated in Table S1 (supplementary data). PLS model was constructed using the calibration set and number of factors in a model was determined by 3-time cross validations (two sample batches for modeling and remained one batch for validation). With using a constructed model, the gemcitabine concentrations of samples in the test set were determined and subsequent accuracy was acquired.

2.3. Determination of gemcitabine concentration of samples using HPLC

Gemcitabine concentrations of 18 samples chosen from the two validation sample batches (9 samples per batch) were also determined using a HPLC (Agilent 1260 HPLC systems (Agilent Technololgies, USA)) equipped with a UV detector at 275 nm and an auto sampler. A USP Packing L7 (C8) column (length: 250 mm, internal diameter: 4.6 mm, grain diameter: 5 μm) was used and the mobile phase consisted of 1.0 L water (pH 2.4) containing 13.8 g NaH_2PO_4 and 2.5 mL H_3PO_4. An isocratic elution was performed at room temperature over 2 min with a flow rate of 1.2 mL/min. Each sample solution was filtered using a PVDF (polyvinylidene difluoride) syringe filter (pore size: 0.45 μm) and the filtered solution was transferred into an autosampler vial (2 mL) for analysis. All of the solvents and reagents were of HPLC grade and purchased from Sigma Aldrich.

3. Results and discussion

3.1. Raman spectra of gemcitabine powder in a glass vial acquired using APO schemes

Fig. 2 (a) and (b) show Raman spectra (1755-255 cm^{-1} range) of pure gemcitabine·HCl, D-mannitol, and sodium acetate, the constituents of gemcitabine injection powder, and the 51.8% gemcitabine powder (not housed in a glass vial). The molecular

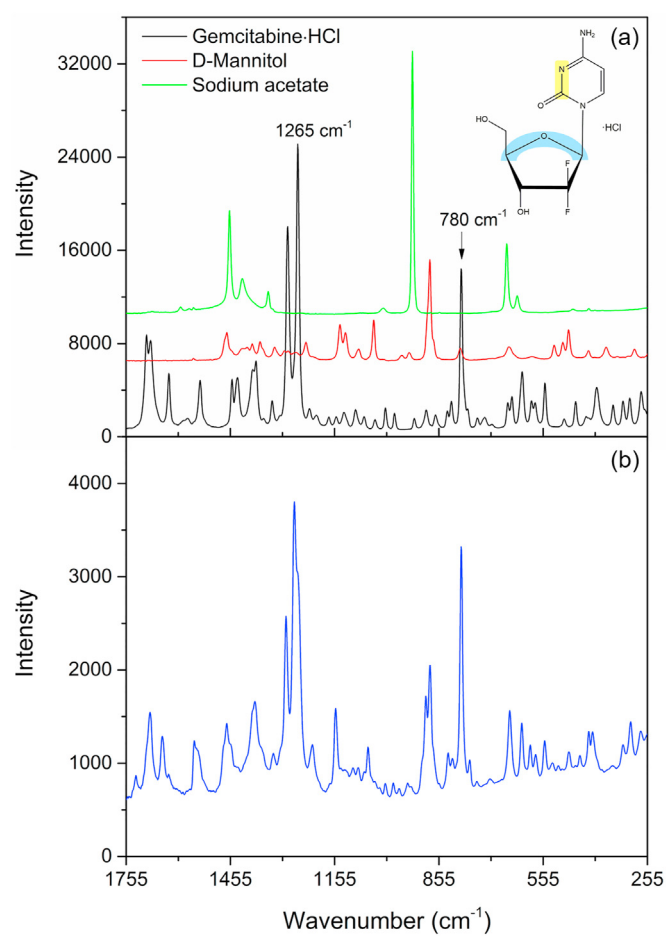

Fig. 2. Raman spectra of gemcitabine·HCl, D-mannitol, and sodium acetate (a), as well as 51.8% gemcitabine injection powder (b). The spectra directly acquired from the samples without housing in a glass vial are shown.

structure of gemcitabine·HCl is also displayed in the figure. Numerous peaks for gemcitabine·HCl are present, and the most intense peak at 1265 cm^{-1} and the second strongest peak at 780 cm^{-1} originate from C—N stretching of the purine ring (designated by the yellow shade on the structure) and C—*O*—C stretching (the blue shade), respectively [3,21]. Further, diverse Raman peaks of mannitol and sodium acetate appear together. The spectrum of 51.8% gemcitabine powder (Fig. 2 (b)) is the result of overlapped features of the components and the feature of gemcitabine is relatively dominant in the 1265 and 780 cm^{-1} peaks due to its high concentration.

Using the APO scheme, Raman spectra of 51.8% gemcitabine injection powder in a glass vial were initially acquired by laser illumination at three different locations on the side of sample-packed glass vial with a fixed position of the detector just beneath the glass vial as shown in Fig. 1 (b). The resulting axial offset distances were 11, 6.5, and 2 mm (referred to as APO-1, APO-2, and APO-3, respectively). The locations of laser illumination in the APO schemes and the corresponding axial offset distances are presented in Fig. 1 (b). Fig. 3 (a) shows the Raman spectra of the 51.8% gemcitabine injection powder in a glass vial acquired using the different APO schemes. The inset magnifies the gemcitabine peak at 780 cm^{-1}. The Raman spectrum of an empty glass container (pink dashed-line) is displayed for comparison. As observed, all three APO schemes produce the glass background in the 1755-909 cm^{-1} range; while, their magnitudes are different. Due to the co-appearing glass background, the gemcitabine peaks in the 1755-909 cm^{-1} range become less characteristic and the strongest 1265 cm^{-1} peak is considerably buried in the background. Meanwhile, the shapes of peaks in the 909-255 cm^{-1} range are clear with no or minimal influence of the glass background.

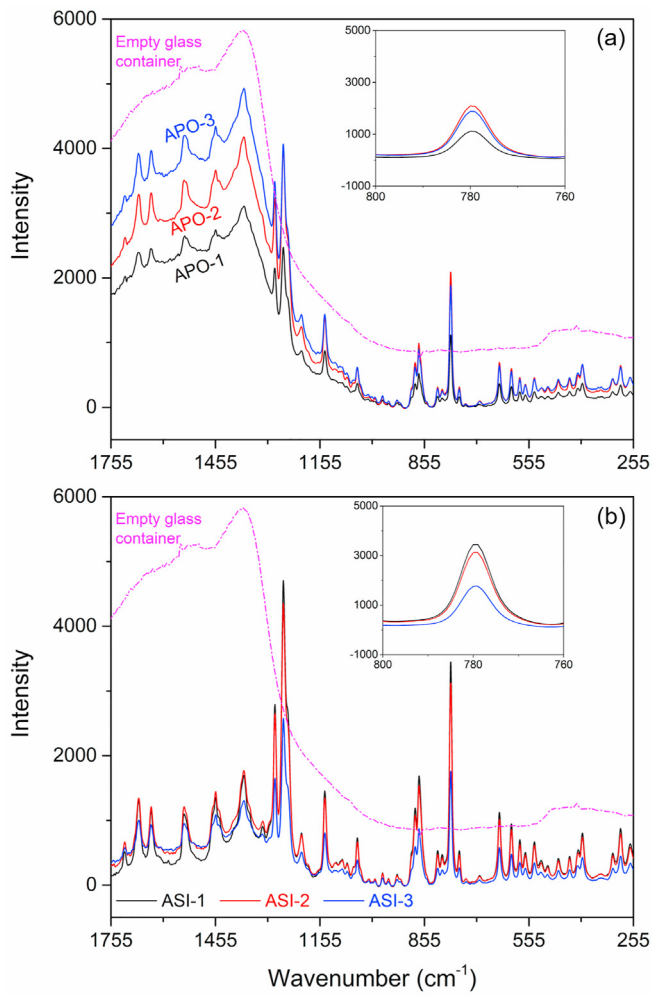

Fig. 3. Raman spectra of 51.8% gemcitabine powder housed in a glass vial acquired using the APO-1, APO-2, and APO-3 schemes (a), and the ASI-1, ASI-2, and ASI-3 schemes (b). In each plot, the gemcitabine peak at 780 cm^{-1} is magnified in the inset and Raman spectrum of an empty glass container (pink dashed-line) is displayed for comparison. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

To quantitatively assess the dominance of the glass background relative to the sample information, the intensity ratio between the gemcitabine peak at 780 cm^{-1} and the glass peak at 1485 cm^{-1} (hereafter denoted as $I_{gemcitabine}/I_{glass}$) in each case was calculated, as shown in Table 1. It is worthwhile to note that no sample peak appears at 1485 cm^{-1}. The absolute intensities of the 780 and 1485 cm^{-1} peaks are also provided in the table. A smaller $I_{gemcitabine}/I_{glass}$ value implies the greater presence of glass background relative to the sample peak. In the case of using the APO-1 scheme providing the longest axial offset distance, the underlying glass background is smallest as expected. Since the generated glass photons at the wall continuously diffuse during their travel inside the sample packing to the detector, a fewer number of the glass photons reaches the detector when the axial offset distance is lengthened. For the same reason, the intensity of the 780 cm^{-1} gemcitabine peak is also smallest (refer to the inset) and the resulting $I_{gemcitabine}/I_{glass}$ is 0.46.

With adopting the APO-2 scheme, the magnitude of glass background increases due to the shortened offset distance and the 780 cm^{-1} gemcitabine peak becomes most intense. The $I_{gemcitabine}/I_{glass}$ rises to 0.68 from the large contribution of the highest gemcitabine peak. The glass background becomes most dominant with the use of the APO-3 scheme due to the shortest offset distance and the corresponding intensity of the sample peak interestingly decreases compared to that in the APO-2 spectrum. Since the laser irradiates on the bottom side of sample packing in the APO-3 scheme, the sampling volume is smaller than that in the APO-2 scheme. This explains the observed decrease of sample peak intensity and the $I_{gemcitabine}/I_{glass}$ drops to 0.49.

Meanwhile, in the previous APO measurement of neat ethanol solution in a glass bottle [12], the intensity of the glass background sensitively decreased by positioning the detector farther from the laser illumination point, and it nearly vanished around the offset distance of 10.0 mm. In contrast, complete removal of the glass background in the above APO measurements is probably difficult due to the broadened diffusion of glass photons by the scattering inside of the packed powder sample. For the examination, the APO-2 scheme possessing the offset distance of 6.5 mm was chosen. The offset distance was lengthened up to 46.5 mm (4 mm interval) and the corresponding intensities at 1485 and 780 cm^{-1} were examined (refer to Fig. 4 (a)). The glass intensity (at 1485 cm^{-1}, red square) gradually decreased until the offset distance reached 30.5 mm and became steady afterward. Even at the offset distance of 46.5 mm, 66% of the glass intensity collected at the 6.5 mm offset remained, thereby confirming the difficulty for complete escape of the glass photons in the detection.

Unlike the variation of the glass background intensity, the decrease of the sample peak (at 780 cm^{-1}, black square) with lengthening the offset distance is rather mild. Two reasons are simultaneously associated to the observation. First, the distribution of generated sample Raman photons is greatly broad due to the continual scattering events into the packing; therefore, they do not significantly vary according to the offset distances. Second, since the size of detector window is large (25.4 mm), the detector can effectively recognize a large portion of the distributed sample photons, although the photon distributions are somewhat different

Table 1
Intensity ratio between the gemcitabine peak at 780 cm^{-1} and glass peak at 1485 cm^{-1} ($I_{gemcitabine}/I_{glass}$), intensity of the glass peak at 1485 cm^{-1}, and intensity of the gemcitabine peak at 780 cm^{-1} in the spectra of 51.8% gemcitabine powder in a glass vial measured using different APO and ASI schemes.

Employed Scheme	$I_{gemcitabine}/I_{glass}$	Glass intensity at 1485 cm^{-1}	Gemcitabine intensity at 780 cm^{-1}
APO-1	0.46	2425.2	1117.4
APO-2	0.68	3079.0	2090.0
APO-3	0.49	3809.4	1881.2
ASI-1	9.20	506.4	4660.0
ASI-2	5.38	579.9	3121.4
ASI-3	2.95	597.4	1760.8

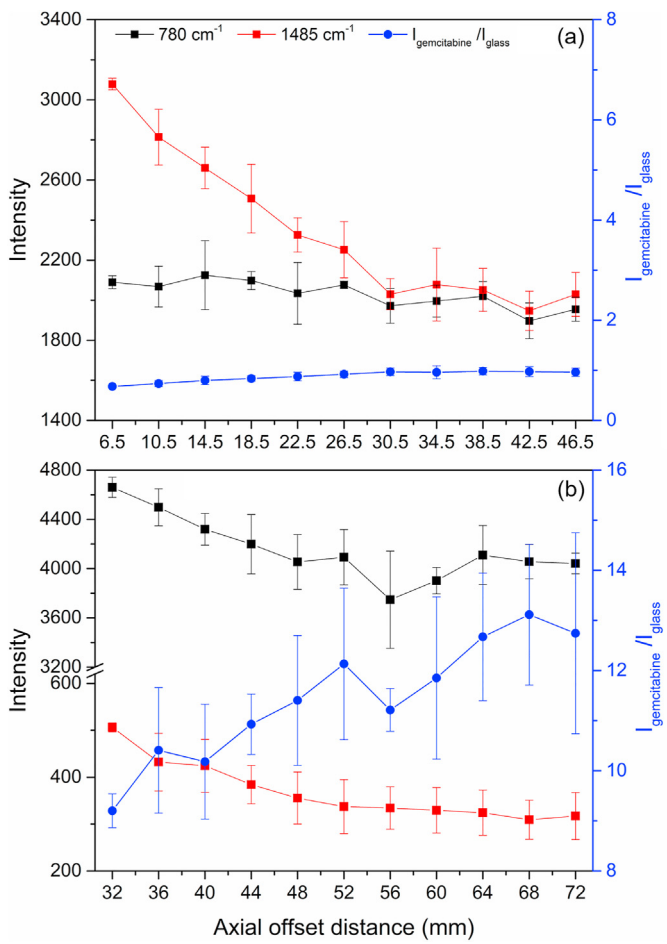

Fig. 4. Variations of the intensity of the glass peak at 1485 cm^{-1}, intensity of gemcitabine peak at 780 cm^{-1}, and $I_{gemcitabine}/I_{glass}$ with changing the offset distance in the APO-2 (a) and ASI-1 (b) schemes up to 46.5 and 72 mm, respectively. Refer to the right blue *y*-scale for the $I_{gemcitabine}/I_{glass}$. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

by the change of the offset distances. In overall, the number of detected photons is not sensitively varied under change of the offset distance. The $I_{gemcitabine}/I_{glass}$ (blue circle) elevates with increasing the offset distance mainly by the decrease of the glass background and starts to level off at the 30.5 mm offset, yielding the $I_{gemcitabine}/I_{glass}$ of 0.97. Meanwhile, the standard deviations (error bars) of the glass and sample peak intensities in the triplicate measurements are small at the 6.5 mm offset, and considerably increase from the 10.5–46.5 mm offset. The larger intensity variations are attributed to the increased fluctuation in the number of detected Raman photons by the lengthened travel distances toward the detector.

3.2. Determination of gemcitabine concentration of samples using APO scheme

The acquired Raman spectra of all 65 samples (five separate batches, 13 samples per batch) housed in a glass vial using the APO-2 scheme with 30.5 mm offset (referred to as the APO-30.5 mm scheme, the $I_{gemcitabine}/I_{glass}$ of 0.97) were employed to determine gemcitabine concentration using PLS. Before PLS, the subtraction of underlying glass background in each spectrum was initially performed [22]. The baseline of each sample spectrum was linearly offset to zero at 909 cm^{-1} and the spectrum of the empty glass vial was linearly scaled to have the same intensity of each sample spectrum at 1485 cm^{-1}. Then, the scaled glass spectrum was subtracted from the corresponding sample spectrum. After subtraction of the glass background, the baseline of the subsequent spectrum was linearly corrected at eight different wavenumbers (1755, 1605, 1333, 1120, 909, 666, 377 and 255 cm^{-1}) and the peak area under the 1755 -255 cm^{-1} was calculated. Finally, each baseline-corrected spectrum was divided by the corresponding peak area for normalization. Fig. 5 (a) shows the glass background-subtracted spectra for all of the samples acquired with the APO-30.5 mm scheme. The top insets magnify the two most intense gemcitabine peaks at 1265 and 780 cm^{-1}. At a glance, the baselines of spectra across the whole spectral range vary in some degree and the variation is larger in the 1755-1155 cm^{-1} range where the glass background is centered. This implies that clear and selective elimination of the underlying glass background in the spectra is difficult especially when the magnitude of the glass background is large. As observed in the insets, the intensities of the 1265 and 780 cm^{-1} gemcitabine peaks vary due to the change of the gemcitabine concentration; while the baselines of these two peaks fluctuate to some extent as mentioned. To examine the variation of peak intensity according to the gemcitabine concentration, the same peaks measured from the 44.6, 49.1, 53.0, and 56.3 wt% gemcitabine powder samples are compared for clear observation (refer to the bottom insets). As seen, the intensities of both peaks increase with the elevation of gemcitabine concentration.

To determine the gemcitabine concentrations of samples using PLS, three different spectral ranges of the 1755-255, 1755-909, and 909-255 cm^{-1}, corresponding to the whole, glass peak-present, and glass peak-absent ranges, respectively, were separately tested. Table 2 shows the resulting standard error of cross-validation (SECV) and standard error of prediction (SEP) in the determination of gemcitabine concentration. In each PLS model, three factors were used. The use of the whole spectral range provided the best result with the SEP of 0.51 wt%. The subsequent repeatability was assessed by calculating relative standard deviation (RSD) of predicted gemcitabine concentrations in the triplicate measurements of each sample. The repeatability was 0.63%, average of the RSDs obtained from the triplicate measurements of each of 26 validation samples. When the glass background-present range (1755-909 cm^{-1}) was employed, the accuracy became worse (SEP: 0.91 wt

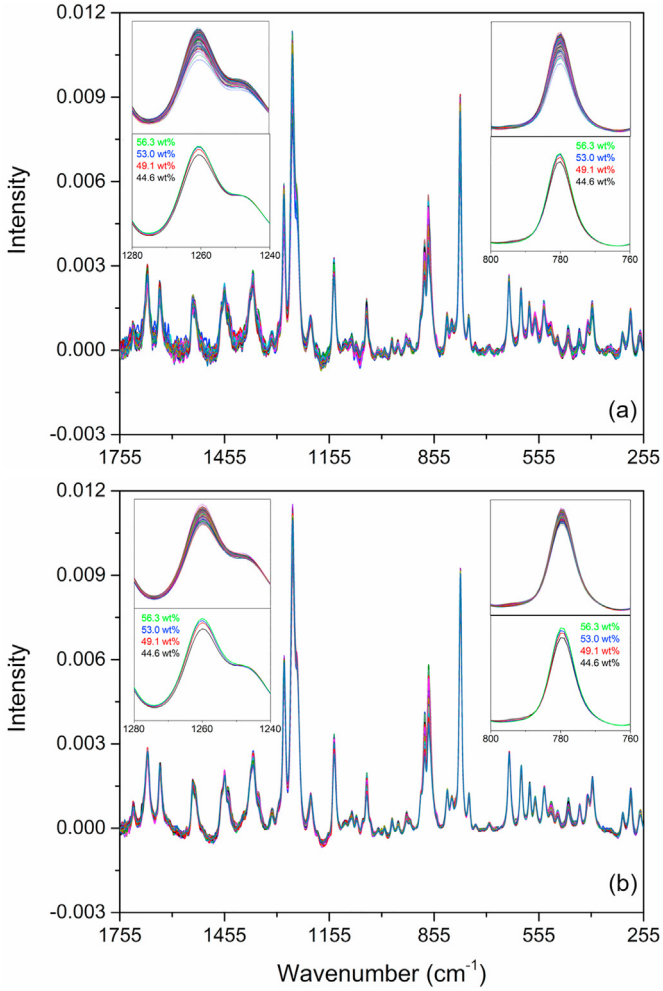

Fig. 5. Glass background-subtracted Raman spectra of all the samples measured using the APO-30.5 mm (a) and ASI-32 mm (b) schemes. The top and bottom insets magnify the 1265 and 780 cm^{-1} gemcitabine peaks of all the samples and 44.6, 49.1, 53.0, and 56.3 wt% samples, respectively.

Table 2
Resulted SECVs and SEPs for the determination of gemcitabine concentration when the spectra collected using the APO-30.5 mm, APO-6.5 mm, and ASI-32 mm schemes are used for PLS.

Unit: wt%	1755-255 cm^{-1}		1755-909 cm^{-1}		909-255 cm^{-1}	
	SECV	SEP	SECV	SEP	SECV	SEP
APO-30.5 mm	0.61	0.51	0.77	0.91	0.54	0.56
APO-6.5 mm	0.36	0.42	0.57	0.65	0.34	0.33
ASI-32 mm	0.21	0.23	0.32	0.38	0.18	0.20

%, repeatability: 1.17%). It clearly shows that the underlying glass background adversely affects the accuracy of the concentration determination, although the background subtraction is performed.

As discussed earlier, the standard deviations of glass and sample peak intensities are smallest at the 6.5 mm offset. The better reproducibility at this condition would be advantageous to secure accuracy, although the underlying glass background in the spectra is larger. Therefore, Raman spectra of the same samples were acquired using the APO-6.5 mm scheme (same as APO-2 scheme in the earlier discussion) and the corresponding spectra were adopted for PLS to determine the gemcitabine concentration. The PLS results are also shown in Table 2. In each PLS model, three factors were also used. In comparison with the results using the APO-30.5 mm scheme, the accuracies are always better using the APO-6.5 mm scheme, and the use of the glass background-free range (909-255 cm^{-1}) enables the most accurate concentration determination (SEP: 0.33 wt%, repeatability: 0.51%). The relevant concentration correlation plot is shown in Fig. 6 (a). The blue circles and red triangles indicate the calibration and validation samples, respectively. The more reproducible sample measurement was the key for the observed accuracy improvement. The regression analysis on the data informed that the standard deviations of residuals, intercept, and slope were 0.41 wt%, 0.425, 0.99, respectively, and R^2 was 0.996.

3.3. Raman spectra of gemcitabine powder housed in glass vial acquired using ASI schemes

Since only partial suppression of the glass background was achievable using the APO scheme, an alternative spectral acquisition scheme able to further diminish the glass background is beneficial for more accurate quantitative analysis. A potential approach is the relocation of the glass photon-generating spot (laser illumination spot) at farther distant from the detector without diminishing intensity of the sample peak. Since empty space exists above the packed sample (Fig. 1), the laser can be oriented at a slant (30° respective to the vertical surface on the glass vial) on the headspace side of vial wall (named as ASI scheme, refer to Fig. 1 (c)). Since the offset distance between the laser illumination and photon detection is increased and generated glass photons need to cross the packed sample in the ASI scheme, fewer glass photons are expected to arrive at the detector.

With this anticipation, Raman spectra of the same 51.8% gemcitabine powder in a glass vial were collected using the ASI schemes with illuminating the laser at three different locations on the side of the glass vial (Fig. 1 (c), referred to as ASI-1, ASI-2, and ASI-3, respectively). The axial offset distance in the ASI-1, ASI-2, and ASI-3 scheme was 32, 25.5, and 19 mm, respectively. The spectra acquired using the three ASI schemes are shown in Fig. 3 (b). As observed, the magnitudes of the glass background in the spectra are much smaller than those in the APO spectra; however, they are not significantly different from one another. The gemcitabine peaks in the 1750-1000 cm^{-1} range become clearly featured, such as the strongest peak at 1265 cm^{-1}. Meanwhile, the intensities of the 780 cm^{-1} sample peak considerably vary as shown in the inset and the ASI-3 scheme produces the least intense sample peak. Since the laser illuminates on the left top of the packed sample in the ASI-3 measurement, some of the laser photons diffuse out of the packing, and subsequently the Raman sampling volume thereby decreases. So, the decrease of the gemcitabine peak is consequential. In the ASI-2 and ASI-1 measurements, the laser illumination spot moves toward the middle top of the sample packing, so the corresponding sample volumes increases. As a result, the sample peak becomes most intense in the ASI-1 measurement and the subsequent $I_{gemcitabine}/I_{glass}$ is greatest, 9.20 (refer to Table 1).

In the ASI-1 measurement, the detector was positioned just beneath the glass vial (offset distance: 32 mm). Similar to the APO measurement, with moving the detector position farther away from the bottom of the glass vial up to 40 mm in the ASI-1 scheme, the intensities at 1485 and 780 cm^{-1} were examined (Fig. 4 (b)). The longest offset distance was 72 mm. As observed, the patterns in the intensity variations of glass and sample peaks are analogous to those in the APO measurement and the $I_{gemcitabine}/I_{glass}$ generally increases with lengthening the offset distance. Nonetheless, the original scheme with the offset distance of 32 mm (hereinafter, referred to as ASI-32 mm) was chosen for subsequent sample

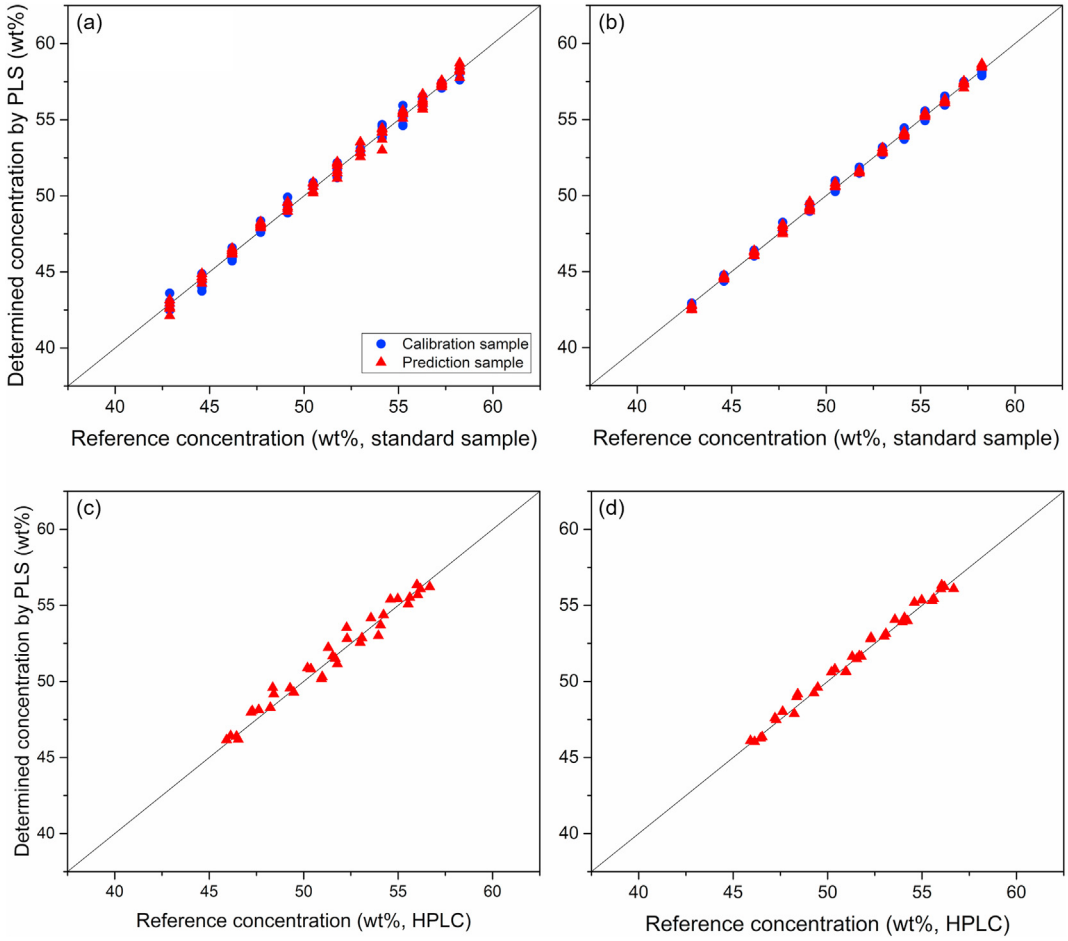

Fig. 6. Resultant concentration correlation plots for the determination of gemcitabine concentration by using the spectra in the 1755-255 cm^{-1} range acquired using the APO-6.5 mm scheme (a) and the spectra in the 909-255 cm^{-1} range collected using the ASI-32 mm scheme (b). The blue circles and red triangles indicate the calibration and validation samples. The correlation plots with respect to the HPLC-determined reference concentrations (*x*-axis) in the APO-6.5 mm (c) and ASI-32 mm (d) measurements are also shown. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

measurement to take advantage of the highest sensitivity (the most intense sample peak) since the magnitude of the glass background is much smaller than that of the gemcitabine peak. Additionally, the standard deviations of glass and sample peak intensities in the triplicate measurements are smallest at this condition.

3.4. Determination of gemcitabine concentration of samples using the ASI scheme

Fig. 5 (b) shows the glass background-subtracted spectra for all of the samples acquired using the ASI-32 mm scheme. Again, the normalized spectra after the baseline correction are shown. The top insets also highlight the two gemcitabine peaks at 1265 and 780 cm^{-1}. At a glance, the baseline variation of the ASI spectra is smaller than that of the APO spectra. The stable baseline is obviously advantageous to secure a more accurate determination of concentration. The bottom insets show both peaks corresponding to the 44.6, 49.1, 53.0, and 56.3% gemcitabine powder samples. The concentration-dependent intensity variations are relative more apparent as seen. Table 2 presents the PLS results using the spectra collected with the ASI-32 mm scheme. In each PLS model, three factors were used again. When the 909-255 cm^{-1} range (glass background-free range) is used for PLS, the SEP decreases to 0.20 wt % and the repeatability also becomes lowered to 0.18%. The corresponding concentration correlation plot is presented in Fig. 6 (b). The predicted concentrations for each sample are more closely gathered along the diagonal line and the triplicate measurements of each sample are considerably less scattered compared to those in the APO measurement. The subsequent standard deviations of residuals, intercept, and slope were 0.19 wt%, −0.540, and 1.01, respectively, and R^2 was 0.999. In comparison with the results using the APO-6.5 mm scheme, the standard deviation of residuals is considerably smaller, alternatively indicating the superior accuracy of ASI measurement; while, the intercept, slope, and R^2 were not noticeably different. Moreover, although the whole spectral range was used, the accuracy was still very good with the SEP of 0.23 wt%. When the glass background-present range (1755-909 cm^{-1}) was employed, the accuracy became worse to the SEP of 0.38 wt%; however, it was still similar to the best accuracy (SEP: 0.33 wt%) achieved in the APO measurement. Overall, the diminished underlying glass background and more reproducible measurement by the ASI-32 mm scheme was the origin for the improved accuracy.

As generally known, HPLC is a conventional and standard method for determination of API concentration in a pharmaceutical sample. Therefore, it is worthwhile to examine correlation between the determined gemcitabine concentrations by HPLC and the proposed Raman measurement. For the examination, 18 samples (concentration range: 46.2−56.3 wt%) were chosen from the two validation sample batches (9 samples per batch) and their gemcitabine concentrations were determined using the HPLC as

described in the experimental section. Fig. 6 (c) and (d) presents the concentration correlation plots resulted in the APO-6.5 mm and ASI-32 mm measurements, respectively, with the HPLC-determined reference concentrations in *x*-axis. The resulting SEPs increase to 0.58 and 0.34 wt% in the APO-6.5 mm and the ASI-32 mm measurements, respectively, compared to those in the same measurements based on the standard sample concentration as the reference (SEPs of 0.33 and 0.20 wt%, respectively). The increased data scatterings (SEPs) along the diagonal line in both cases are acceptable since the HPLC analysis inevitably has own uncertainty in the determination of component concentration. Nonetheless, the concentration correlation in the ASI-32 mm measurement is still superior with the mean percent error of 0.54%. This result demonstrates the ASI scheme is a comparable analytical method to HPLC and thereby field-applicable as an alternative non-destructive analysis tool.

Since the gemcitabine peaks in the sample spectra were intense and easily recognizable due to the high concentrations (42.9–58.2 wt%), the adverse influence of underlying glass background on the accuracy of concentration determination would be relatively small and the performance of ASI-32 mm scheme is thereby superior. Meanwhile, if the ASI or APO schemes are used to measure low concentration of content samples such as below 3%, the glass background more seriously hamper the identification of low intensity analyte peaks [9] and subsequent accuracy expects to degrade [23,24]. Under this situation, more sophisticated experimental and/or chemometrics schemes able to overcome the anticipating degradation become necessary. This is a worthwhile and interesting issue to be studied for further exploration of the ASI scheme.

4. Conclusion

For direct Raman measurement of the gemcitabine injection powder samples housed in a glass container without manual sampling, the demonstrated ASI scheme was effective to reduce the co-presenting glass background in the sample spectra and enabled a more reproducible measurement. The accurate determination of gemcitabine concentration was thereby the consequence. Since partially-filled samples in a glass container are frequently encountered in practical applications, the ASI scheme could be a potentially fast and non-sampling Raman spectroscopic analysis tool. It is reasonable to expect that optimal positions of laser illumination and photon detection in ASI scheme vary depending on diverse factors such as the size of a glass container, container wall thickness, and the physical properties of the sample. Therefore, evaluation on the inter-relationship among the mentioned parameters with regard to accuracy of ASI-based measurement is of great interest and thereby underway in this research group. Also, the performance of ASI scheme in measurement of low API content samples will be evaluated.

CRediT authorship contribution statement

Yoonjeong Lee: Formal analysis, Data curation. **Jaejin Kim:** Methodology, Resources. **Janghee Han:** Resources. **Haeseong Jeong:** Investigation. **Young-Ah Woo:** Supervision, Writing – review & editing. **Hoeil Chung:** Conceptualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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